Introduction to Deep Learning

Session 7: Generalization & Regularization

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Lecture Overview

1. Generalization

2. Generalization Paradox

3. Regularisation Techniques

Where We Are

So far we have:

- Network $f_{\boldsymbol{\theta}} : \mathcal{X} \to \mathcal{Y}$ - parameterized function
- Loss $\mathcal{L}(\boldsymbol{\theta}; \mathcal{D})$ - measures fit on training data
- Optimization - algorithms to minimize $\mathcal{L}(\boldsymbol{\theta}; \mathcal{D})$

Open question:

Does minimizing training loss give us a **good model**?

This lecture: what good actually means, and how to get there

True Goal

What we optimize:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \ell(f_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), y^{(i)})$$

What we actually want: low loss on **new, unseen examples from the population**

$$\mathcal{L}_{\text{pop}}(\boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{x},y)\sim p_{\text{data}}}\left[\ell(f_{\boldsymbol{\theta}}(\mathbf{x}), y)\right]$$

Generalization gap:

$$\underbrace{\mathcal{L}_{\text{pop}}(\boldsymbol{\theta})}_{\text{true goal}} - \underbrace{\mathcal{L}_{\text{train}}(\boldsymbol{\theta})}_{\text{what we minimize}} \geq 0$$

Test set: finite sample we create to *estimate* $\mathcal{L}_{\text{pop}}$ - **measurement tool, not the goal**!

Train / Validation / Test Split

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Training set

Fit model parameters

$$\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}; \mathcal{D}_{\text{train}})$$

Validation set

Select model, tune hyperparameters

$$\mathcal{L}(\boldsymbol{\theta}; \mathcal{D}_{\text{val}})$$

Touched many times during development

Test set

Final honest evaluation

$$\mathcal{L}(\boldsymbol{\theta}; \mathcal{D}_{\text{test}})$$

Touched once, at the end

Common mistake: tuning on test set - reported performance is optimistic

Typical splits: 80/10/10 or 70/15/15 - depends on dataset size

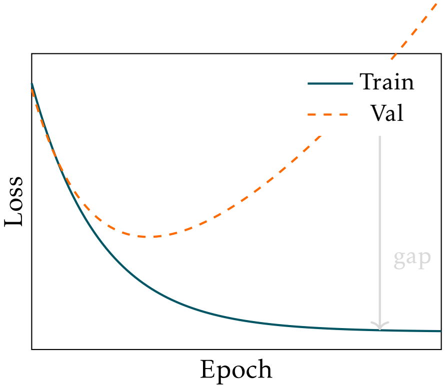
Overfitting and Underfitting

Underfitting: model too simple

- High train loss, high test loss
- Small generalisation gap

Overfitting: model memorises training data

- Low train loss, high test loss
- Large generalisation gap



Signal: monitor generalisation gap - **train loss not enough for generalisation**

What To Do

Underfitting (high train loss, small gap)

- Increase model capacity
- Train longer
- Tune learning rate and optimiser
- Reduce regularisation if applied

Overfitting (low train loss, large gap)

- More data - always most effective
- Regularisation - this lecture
- Reduce model capacity

Bias-Variance Tradeoff

Classical view: test error decomposes into two competing terms

Bias - error from wrong assumptions

- Model too simple to capture the data
- High train loss, high test loss
- Underfitting

Variance - sensitivity to training set

- Model fits noise in training data
- Low train loss, high test loss
- Overfitting

Classical prescription: choose model complexity to balance the two

More parameters ⇒ lower bias, higher variance ⇒ overfitting

Limits of Classical View

Classical prediction: more parameters than training examples ⇒ catastrophic overfitting

Reality in deep learning:

- Deep NNs: huge numbers of params - can memorise training data
- Deep NNs: routinely drive train loss close to zero
- Classical theory: model that nearly memorises train data ***cannot generalise***
- Deep NNs: ***generalise well in practice*** ... hmm ...

Why do deep NNs generalise despite being massively overparameterised?

Generalization Paradox

Many Good Minima

Classical concern: gradient descent finds a local minimum - but which one?

Empirical observation in deep networks:

- Overparameterised networks - vast number of local minima
- Most local minima have very similar loss values
- Different intialisation -> different minima -> ***yet generalise similarly***

Why: NNs far more params than needed to fit the data.

Many equivalent solutions exist - which one we find not so critical

Flat vs Sharp Minima

Not all minima generalise equally well

Sharp minimum:

- Loss changes rapidly
- Small distribution shift $\Rightarrow$ loss jumps
- Poor generalisation

Flat minimum:

- Loss changes slowly
- Small distribution shift $\Rightarrow$ loss stays low
- Good generalisation

SGD: mini-batching stochastic noise - favours flat minima -> implicit regularisation

Double Descent

Classical bias-variance curve: test error has one optimal complexity - U-shaped curve

Double descent: beyond the interpolation threshold, test error decreases again

- **Underfitting:** too little capacity
- **Interpolation threshold:** maximally rigid - worst generalisation
- **Overparameterised:** smoother interpolation - generalisation recovers

Deep networks operate in overparameterised regime

Regularisation Techniques